

Analysis Comprehensive Examination

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Location: Education 338

Instructions

This examination consists of three parts labeled A, B, and C.

- Part A covers the Core Material. It consists of 7 questions for a total possible score of 40 marks. All questions in Part A are mandatory.
- Part B covers the specialized material on Computational Mathematics. This section consists of 3 questions worth 15 marks each, of which you must attempt 2, for a total possible score of 30 marks.
- Part C covers the specialized material on Differential Equations. This section consists of 3 questions worth 15 marks each, of which you must attempt 2, for a total possible score of 30 marks.

Please take note of the following:

1. In Parts B and C, if you attempt more than 2 questions, you must clearly indicate which 2 questions you would like marked. Otherwise, only the first 2 answered questions in the order they appear in your solutions will be graded.
2. The values of individual questions in Part A may not be the same.
3. The examination is worth a total of 100 marks. You need at least 75 marks to pass.
4. The total time of the examination is six hours. No texts, reference books, calculators, cell phones, smart watches, or other aids are permitted during the examination.

Part A: Core material

1. (5 points) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is twice differentiable everywhere. Using the definition of bounded variation, prove that f has bounded variation on any closed interval $[a, b]$.
2. (4 points) Suppose f is Riemann integrable on $[a, b]$, and $f(x) > 0$ for every $x \in [a, b]$. Prove that $\int_a^b f(x) dx > 0$, where the integral is understood as a Riemann integral.
3. (6 points) Let $C([0, 1])$ be the space of continuous real-valued functions on $[0, 1]$ with the supremum norm $\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|$. Let K be a constant and $0 < \alpha \leq 1$. Define the set $\mathcal{F} \subset C([0, 1])$ as:

$$\mathcal{F} = \{f \in C([0, 1]) : |f(0)| \leq M \text{ and } |f(x) - f(y)| \leq K|x - y|^\alpha \text{ for all } x, y \in [0, 1]\}$$

Prove that $\mathcal{F}$ is a compact subset of $C([0, 1])$.

4. Suppose a 2π -periodic function is defined by

$$f(x) = \begin{cases} 1, & x \in [0, \pi), \\ -1, & x \in [-\pi, 0). \end{cases}$$

- (a) (4 points) Find the Fourier series of f with respect to the standard trigonometric system.
 - (b) (2 points) What is the pointwise sum of this Fourier series (if it exists)?
 - (c) (3 points) Does the series converge uniformly on $\mathbb{R}$?
5. (5 points) Evaluate the following limit and justify your steps:

$$\lim_{n \rightarrow \infty} \int_0^\infty \frac{n \sin(x/n)}{x(1+x^2)} dx$$

6. (5 points) Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. Suppose $f : \mathbb{D} \rightarrow \mathbb{D}$ is analytic and has at least two distinct fixed points $z_1, z_2 \in \mathbb{D}$, i.e. $f(z_i) = z_i$, $i = 1, 2$. Prove that $f(z) = z$ for all $z \in \mathbb{D}$. Hint: consider an automorphism of $\mathbb{D}$ mapping z_1 to 0.
7. (6 points) Use methods of complex analysis to evaluate

$$\int_{\mathbb{R}} \frac{1 - \cos x}{x^2} dx.$$

Part B: Computational Mathematics

Solve 2 out of the following 3 problems. You must clearly indicate which 2 questions you would like marked. Otherwise, only the first 2 answered questions in the order they appear in your solutions will be graded.

1. Let $\{p_n(x)\}_{n=0}^{\infty}$ be a family of orthogonal polynomials with respect to the inner product $\langle f, g \rangle = \int_a^b f(x)g(x)w(x)dx$ for some positive integrable weight $w(x)$.

(a) (4 points) Show that for $n > 0$ there exists a so-called three-term relation:

$$xp_n(x) = a_np_{n+1}(x) + b_np_n(x) + c_np_{n-1}(x).$$

(b) (2 points) If $p_n(x)$ are orthonormal, show that $c_n = a_{n-1}$.

(c) (2 points) If $a = -b$ and if $w(x)$ is even, show that $b_n = 0$.

(d) (7 points) Suppose that $U_n(x) = \frac{\sin[(n+1)\cos^{-1}(x)]}{\sqrt{1-x^2}}$ are orthogonal polynomials on $(-1, 1)$ with respect to the weight $w(x) = 2\sqrt{1-x^2}/\pi$. Compute the corresponding three-term relation.

2. Tridiagonal divide-and-conquer eigenvalue algorithms are based on the following.

(a) (5 points) Let $D = \text{diag}(d_1, \dots, d_n)$ be a diagonal matrix with $d_i < d_j$ for $1 \leq i < j \leq n$, and let the vector $z \in \mathbb{R}^n$ satisfy $z_i \neq 0$ for all $1 \leq i \leq n$. Show that the eigenvalues of $A = D + zz^\top$ are the zeros of the rational function:

$$f(\lambda) = 1 + \sum_{i=1}^n \frac{z_i^2}{d_i - \lambda}.$$

(b) (5 points) Sketch the graph of $f(\lambda)$ indicating all horizontal and vertical asymptotes. Describe n disjoint open bounded intervals each containing exactly one eigenvalue.

(c) (5 points) Describe an iterative method that is guaranteed to converge to the eigenvalues of A . Estimate the number of floating-point operations required as a function of n and relative accuracy $\epsilon > 0$.

3. (a) (5 points) Consider the matrices A and $B \in \mathbb{R}^{n \times n}$, where:

$$A = \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{bmatrix}.$$

Use the Cholesky factorization to determine if the matrices are positive-definite.

- (b) (10 points) Describe a second order finite difference scheme for the Poisson equation with Neumann boundary condition on the square $\Omega = (0, 1)^2$:

$$\begin{aligned} -\Delta u &= f, \quad \text{for } (x, y) \in \Omega, \\ \frac{\partial u}{\partial n} &= 0, \quad \text{for } (x, y) \in \partial\Omega, \end{aligned}$$

using an offset grid. Include any conditions on f and u to ensure your scheme produces a unique solution.

Part C: Differential Equations

Solve 2 out of the following 3 problems. You must clearly indicate which 2 questions you would like marked. Otherwise, only the first 2 answered questions in the order they appear in your solutions will be graded.

1. Let $T > 0, x_0 \in \mathbb{R}$, and consider $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$. This question is concerned with the following differential equation

$$(1) \quad \frac{dx}{dt} = f(t, x(t)), \quad t \in (0, T), \quad x(0) = x_0,$$

where the unknown is the function $x = x(t)$ for $t \in [0, T]$.

- (a) (3 points) Suppose $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ is continuous and that there exists $L > 0$ such that

$$|f(t, x) - f(t, y)| \leq L|x - y|, \quad \forall t \in [0, T], \forall x, y \in \mathbb{R}.$$

For every $x \in C([0, T])$, we define Px to be the function which takes the values

$$Px(t) := x_0 + \int_0^t f(s, x(s)) \, ds, \quad \forall t \in [0, T].$$

Prove that $P : C([0, T]) \rightarrow C([0, T])$ is well-defined and satisfies the estimate

$$|Px(t) - Py(t)| \leq tL \sup_{s \in [0, T]} |x(s) - y(s)|, \quad \forall t \in [0, T], \forall x, y \in C([0, T]).$$

- (b) (7 points) Let f be as in the previous part. Prove that (1) has a unique solution $x \in C^1((0, T)) \cap C([0, T])$. *Hint:* Use the previous part. You may assume that $C([0, T])$, endowed with the sup norm / metric, is a non-empty complete metric space.
- (c) Let $x_0 = 0$ and consider

$$f(t, x) = \begin{cases} x^{\frac{1}{2}} & \text{if } x \geq 0 \\ -(-x)^{\frac{1}{2}} & \text{if } x < 0 \end{cases}, \quad \forall t, x \in \mathbb{R}.$$

In particular, f is independent of t .

- (ci) (2 points) For any $T > 0$, explicitly demonstrate an uncountably infinite number of solutions to (1).
- (cii) (3 points) Explain why the non-uniqueness of solutions in this example does not contradict the previous result.

2. Denote by B the unit ball in $\mathbb{R}^N$. Let $f \in L^2(B)$ and consider the Neumann problem for the following elliptic equation:

$$(2) \quad \begin{cases} -\Delta u + u = f & \text{in } B \\ \frac{\partial u}{\partial n} = 0 & \text{on } \partial B \end{cases},$$

where n is the unit outward normal for B i.e. $n(x) = x$ for $x \in \partial B$.

- (a) (5 points) Let $A : H^1(B) \times H^1(B) \rightarrow \mathbb{R}$ be defined by

$$A(u, v) := \int_B u(x)v(x) + \nabla u(x) \cdot \nabla v(x) \, dx, \quad \forall u, v \in H^1(B).$$

We say that $u \in H^1(B)$ is a *weak solution* to (2) if

$$A(u, v) = \int_B f(x)v(x) \, dx, \quad \forall v \in H^1(B).$$

Show that if $u \in H^2(B)$ is a pointwise solution to (2), then it is also a weak solution to (2). *Note:* You should comment on the presence / absence of the Neumann boundary condition in the weak solution formulation.

- (b) (10 points) Prove that there exists a unique weak solution $u \in H^1(B)$ to (2). Moreover, show that $\|u\|_{H^1(B)} \leq \|f\|_{L^2(B)}$.

3. Let $\Omega \subset \mathbb{R}^N$ be open, bounded, and connected.

- (a) (5 points) Suppose $u \in C^2(\Omega)$ is harmonic (in Ω). Show that, for any $x \in \Omega, r > 0$ such that $B_r(x) \subset \Omega$, we have

$$(3) \quad u(x) = \oint_{\partial B_r(x)} u(y) \, dS(y) = \oint_{B_r(x)} u(y) \, dy.$$

Here, $\oint$ denotes the *average integral* and $dS(y)$ denotes the (boundary) surface measure for $y \in \partial B_r(x)$.

- (b) (5 points) Suppose $u \in C^2(\Omega)$ satisfies (3) for every $x \in \Omega, r > 0$ such that $B_r(x) \subset \Omega$. Prove that u is harmonic (in Ω).
- (c) (5 points) Suppose $u \in C^2(\Omega) \cap C(\overline{\Omega})$ is harmonic (in Ω). Assume there is some point $x_0 \in \Omega$ such that

$$u(x_0) = \max_{x \in \overline{\Omega}} u(x).$$

Prove that u must be constant within Ω .